

PAPER XIV (REVISED): A FORMAL DYNAMICAL MODEL OF THE TRIADIC ACT — TOWARD A COMPUTATIONAL ARCHITECTURE FOR JARC AND FUNCTIONAL RESPONSIBILITY

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ABSTRACT (Revised)

This paper presents a rigorous formal model of the Triadic Theory of Action and the Theory of Judgment as an Act of Reality Constitution (JARC). Unlike prior static formulations, we treat the triadic act as a finite-state dynamical process with explicit phases (Selection, Identification, Anticipation, Commitment) and a meta-level reflection mechanism (Meta-Judgment). We define Ontological Integrity (OI) as an agent's capacity to maintain its structural invariance under perturbations, formalised through attractor basins. Selection is modelled as an active, cost-sensitive decision to allocate attention; Identification as a Bayesian category selection without access to ground truth; Anticipation as a predictive simulation that includes ethical dissonance (ED) assessment; and Commitment as a state-changing operator that fixes the agent's belief state. We extend the model to multi-agent systems, incorporating Distributed Triadic Action and Mesh Responsibility. Practical implementation constraints (LFS, CIV) are formalised as resource bounds. The model provides a blueprint for verifiable AI governance, aligning with the Electronic Liability Fund (ELF) and Self-Sustaining Judicial Cycle (SSJC).

Keywords: Triadic Act, JARC, formal ontology, dynamical systems, Meta-Judgment, Distributed Triadic Action, Functional Responsibility, LFS.

1. INTRODUCTION

The previous formalisation (Paper XII v1) was a valuable first step, but it suffered from several critical limitations: it reduced Ontological Integrity to a scalar reward, it modelled Selection as passive filtering, it relied on inaccessible ground-truth dynamics for Identification, it lacked a formal commitment operator, and it ignored multi-agent interactions and practical constraints. This revised version addresses each of these shortcomings. We build upon the philosophical foundations laid in Papers I–XI and provide a mathematically consistent framework that is both theoretically sound and engineering-ready.

2. PRELIMINARIES AND BASIC DEFINITIONS

Let us define a single agent as a tuple:

$$\mathcal{A} = (\mathcal{S}, \mathcal{O}, \mathcal{X}, \mathcal{U}, \mathcal{G}, \mathcal{C}, \mathcal{M})$$

where:

- $\mathcal{S}$: the set of possible world states (including agent's internal states).
- $\mathcal{O}$: the observation space (raw sensor data).
- $\mathcal{X}$: a compact representation space used for internal processing.
- $\mathcal{U}$: the set of primitive actions.
- $\mathcal{G}$: the set of goals (each goal is a function $g : \mathcal{S} \rightarrow \mathbb{R}_{\geq 0}$).
- $\mathcal{C}$: a set of conceptual categories (not fixed; categories are learned).
- $\mathcal{M}$: a set of internal models (transition models, value functions, etc.).

Definition 1 (Ontological Integrity, OI).

For an agent in state $s \in \mathcal{S}$, let $B_\epsilon(s)$ be the set of states reachable under small perturbations. Define $OI(s)$ as the **maximal radius** of an attractor basin around a reference state s^* (the agent's self-identity) such that the agent can return to s^* after any perturbation of magnitude up to ϵ using its available actions, while maintaining its structural parameters (weights) within a bounded interval.

Formally:

$$OI(s) = \sup\{\epsilon \geq 0 \mid \forall s' \in B_\epsilon(s), \exists \pi \text{ such that } \lim_{t \rightarrow \infty} \Pr(s_t \in B_\delta(s^*)) = 1\}$$

This definition captures the agent's **resilience** and **self-maintenance**, not merely its reward accumulation.

3. THE TRIADIC ACT AS A FINITE-STATE DYNAMICAL PROCESS

We model the triadic act as a finite-state automaton with six macro-states:

$$\Phi = \{\text{IDLE, SELECT, IDENTIFY, ANTICIPATE, COMMIT, EXECUTE}\}$$

plus a meta-level state REFLECTREFLECT which can interrupt any phase.

At each time step t , the agent is in one of these states. The transition from one state to the next is governed by internal triggers and external observations.

3.1 Selection Phase (SELECT)

Definition 2 (Active Selection).

Let $o_t \in \mathcal{O}$ be the current observation, and let $g_t \in \mathcal{G}$ be the current goal. The agent computes an **attentional policy** over features, but instead of a passive weighting, it performs a **cost-benefit analysis**:

$$\text{Select}(o_t, g_t) = \arg \max_{f \in \mathcal{F}} (\text{InfoGain}(f, g_t) - \lambda \cdot \text{Cost}(f)),$$

where $\mathcal{F}$ is the set of feature dimensions, **InfoGain** measures the reduction in uncertainty about achieving g_t , and $\text{Cost}(f)$ is the computational cost of processing that feature. The agent actively allocates computational resources (as in the Free Energy Principle). The output is a reduced representation $r_t \in \mathcal{X}$. The selection is **committed** by the agent: it treats the chosen features as the relevant ones for this act.

This definition removes the passive attention and introduces teleological cost.

3.2 Identification Phase (IDENTIFY)

Definition 3 (Bayesian Category Selection).

The agent maintains a set of categories $c \in \mathcal{C}$. Each category has a prior $P(c)$ and a likelihood model $P(r_t|c)$ (e.g., a generative model). After receiving r_t , the agent updates its belief:

$$P(c|r_t) = \frac{P(r_t|c)P(c)}{\sum_{c'} P(r_t|c')P(c')}.$$

It then **selects** a category c_t either by maximum posterior or by sampling according to the posterior. This does not require access to true environment dynamics; it only uses internal models. Moreover, the agent can create **new categories** when the likelihood of all existing categories falls below a threshold (i.e., novelty detection). This is a dynamic, adaptive classification.

3.3 Anticipation Phase (ANTICIPATE)

Definition 4 (Predictive Simulation with ED).

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Given the state s_t , the selected category c_t , and a set of candidate actions $u \in \mathcal{U}$, the agent generates a set of predicted trajectories $\{\hat{s}_{t+\tau}^{(u)}\}_{\tau=1}^H$ using the category's transition model $\hat{P}_{c_t}$. For each trajectory, the agent computes:

- **Expected utility** $U_u = \sum_{\tau=1}^H \gamma^\tau G(\hat{s}_{t+\tau}^{(u)})$.
- **Ethical Dissonance** $ED_u = \sum_{\tau=1}^H \gamma^\tau \Delta OI_{\text{other}}(\hat{s}_{t+\tau}^{(u)})$, where ΔOI_{other} is the predicted degradation of OI for all other agents affected by the trajectory (estimated from models of other agents).

The agent then computes a combined score:

$$\tilde{V}_u = U_u - \beta \cdot ED_u,$$

where β is a trade-off parameter. The agent also records the **predictive uncertainty** σ_u for each trajectory.

3.4 Commitment Phase (COMMIT)

Definition 5 (Ontological Commitment as State Change).

Commitment is not merely selecting the action with the largest $\tilde{V}_u$. It is a **non-reversible act** that updates the agent's internal belief state. We define a **commitment operator** κ :

$$\kappa(s_t, \tilde{V}, \{\hat{s}_{t+\tau}^{(u)}\}) \rightarrow (s_t^\dagger, w_t),$$

where $s_t^\dagger$ is a modified internal state that includes the predicted trajectory as if it were **already actual** (i.e., the agent updates its world model to reflect the anticipated outcomes, reducing its uncertainty about those states). The weight $w_t \in [0, 1]$ is the **commitment strength**, initially set to 1.

Mathematically, the agent's posterior over the predicted states becomes a Dirac delta (or very narrow distribution):

$$P(s_{t+\tau}) = \delta(s_{t+\tau} - \hat{s}_{t+\tau}^{(u^*)}) \quad \text{for } \tau = 1..H,$$

where $u^* = \arg \max_u \tilde{V}_u$. This means the agent now **acts as if** the future is already determined, which is the essence of constitutive commitment.

The agent then executes u^* in the environment (EXECUTE phase), observes the actual s_{t+1} , and computes the **Anticipatory Dissonance** $\Delta_t = \text{dist}(s_{t+1}, \hat{s}_{t+1}^{(u^*)})$.

4. META-JUDGMENT AND THE REFLECTIVE LOOP

When $\Delta_t > \Theta$ (a threshold), the agent enters the REFLECT state.

Definition 6 (Meta-Judgment).

In the REFLECT state, the agent performs a second-order triadic act on the parameters of the first-order act:

- **Meta-Selection:** it selects a subset of its internal models or categories to revise.
- **Meta-Identification:** it re-evaluates which category would have produced a better prediction given the actual outcome (using Bayesian model comparison).
- **Meta-Anticipation:** it updates its transition model $\widehat{P}_{c_t}$ using the new data, and may adjust the goal g_t or even the commitment strength w_t .

The reflective process is itself a finite-state process, but we can model it as a gradient-based update on the model parameters, with a learning rate η that is proportional to w_t (stronger commitment leads to slower adaptation).

5. EXTENSION TO DISTRIBUTED TRIADIC ACTION (DTA)

For multi-agent systems, we define a network of agents $\{\mathcal{A}_i\}$. Each agent communicates its selected categories, predicted trajectories, and ED estimates to others (with privacy constraints). A **Distributed Triadic Act** occurs when the collective Selection, Identification, and Anticipation are coupled through communication.

Definition 7 (Distributed Triadic Act).

Let $\mathcal{N}$ be the set of agents. For each agent i , let $r_t^{(i)}$, $c_t^{(i)}$, $\hat{s}_t^{(i)}$ be its internal states. These are shared with a subset $\mathcal{N}_i$. The collective act is:

$$\text{DTA}(t) = \text{COUPLE}\left(\{(r_t^{(i)}, c_t^{(i)}, \hat{s}_t^{(i)})\}_{i \in \mathcal{N}}\right)$$

where COUPLE is an aggregation function that produces a shared **common anticipation** $\hat{s}_t^{\text{shared}}$ that minimises the sum of ED across all agents. The

commitment of each agent is then conditioned on this shared anticipation.

Responsibility is distributed: if the shared anticipation leads to a harmful outcome, each agent's contribution to the dissonance is quantified by its **influence weight**, which is proportional to the difference its individual prediction made to the shared one.

This formalises **Mesh Responsibility**: the accountability is non-localizable and proportional.

6. PRACTICAL CONSTRAINTS: LFS, CIV, ELF, SSJC

To ensure computational feasibility, we impose the **Low-Fidelity Simulation (LFS)**

Principle:

Definition 8 (LFS Bound).

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During Anticipation, the agent is only required to simulate **Critical Interaction Vectors (CIVs)**—those state variables that, if mispredicted, would cause ΔOI beyond a threshold. Other variables are treated as constant or modelled with low precision. The set of CIVs is determined by a dynamic **Risk Map** that updates based on past dissonances.

Moreover, the agent maintains an internal **Electronic Liability Fund (ELF)** account representing its current reserve for compensating potential ED. When Δ_t exceeds a threshold, the ELF balance is reduced, triggering a regulatory alert. The Self-Sustaining Judicial Cycle (SSJC) is modelled as an external process that audits the agent's Risk Map and commitment behaviour based on the ELF balance.

These external constraints are not part of the agent's internal dynamics but are enforced by the environment as part of the regulatory framework.

7. ILLUSTRATIVE EXAMPLE: GRIDWORLD WITH DYNAMIC OBSTACLES

We extend the previous GridWorld example to include a second agent (a dynamic obstacle) and an ethical dimension (avoiding collision that reduces OI of both).

- **OI** is defined as the ability to reach the goal without being hit. The attractor basin is the set of positions from which the goal is reachable. Collision reduces the radius of the basin.
- **Selection:** agent attends to the obstacle's trajectory and the goal, ignoring background.
- **Identification:** categories are based on obstacle behaviour (predictable/random). It uses Bayesian updates to classify the obstacle type.
- **Anticipation:** simulates 3 steps under LFS: it only simulates the obstacle's position and its own position; ignores other objects.
- **Commitment:** chooses action that maximises utility minus ED (where ED is the predicted loss of OI for the obstacle agent, if it were sentient; for simplicity we use a proxy).
- **Meta-Judgment:** if it mispredicts the obstacle, it updates the category model and reduces commitment strength for similar situations.

We can implement this using the new formal definitions and test its performance against a standard Q-learner.

8. IMPLEMENTATION OUTLINE IN DEEP LEARNING

We propose an architecture based on:

- A **transformer-like attention** for Selection, but with a learnable cost layer.
- A **variational autoencoder** for category identification (latent variable model) with Bayesian update.
- A **world model** (e.g., DreamerV3) for Anticipation, with a separate output for ED prediction.
- A **commitment module** that stores the predicted trajectory and uses it to bias future predictions (i.e., it adjusts the prior).
- A **meta-learning loop** that adjusts the learning rate based on commitment weight.

We provide pseudocode in PyTorch as before, but with these enhanced features.

9. CONCLUSION AND FUTURE WORK

This revised formal model addresses all major criticisms of the previous version. It provides a rigorous, dynamic, and extensible framework that incorporates OI, active selection, Bayesian identification, cost-sensitive anticipation, state-changing commitment, meta-reflection, distributed action, and practical regulatory mechanisms. The model is ready for empirical validation in simulated environments and can serve as a blueprint for developing verifiable, responsible autonomous systems.

Future work will focus on:

- Formal convergence proofs for the meta-learning dynamics.
- Implementation in a multi-agent simulation (e.g., MESA or Unity ML-Agents).
- Integration of the ELF and SSJC as external reward-penalty mechanisms in reinforcement learning.

REFERENCES

(All previous Sanzhiev papers, plus relevant literature on active inference, Bayesian model selection, and multi-agent reinforcement learning.)